

EasyService 20-Slide Presentation Deck

Slide 1: EasyService Architecture & Implementation Overview

Cloud-Native Enterprise Marketplace Platform — Group Two

- Course: Cloud Computing & Security (Microservices Architecture)
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 - Lead Contributor: Abel Seleshe (Architecture & Decomposition Lead)
 - Team Members: Baheran Tesfaye, Nebiyu Yohannes, Wondesen Teshale, Esrom Basazinaw, Yohannes Seyum
 - Target Implementation: Java 21, Spring Boot 3.2, Svelte Frontend & Docker/K8s
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Slide 2: Executive Summary & System Vision

Transforming Monolithic Service Portals into Autonomous Microservices

- Vision: Modern Ethiopian marketplace unifying Hotels, Mobility, Crafts, and Local Services
 - Core Problem: Monolithic bottlenecks, shared database contention, and unified blast radiuses
 - Architectural Solution: 6 Autonomous Bounded Context Microservices
 - Security Foundation: Stateless OAuth2/JWT incorporating Fayda National Digital ID claims
 - Key Capabilities: Reactive Wallet, Dynamic Spin-Wheel Engine & 1-Click Hot Deals Booking
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Slide 3: EasyService Monolith Decomposition

Decomposing Legacy Systems into 6 Bounded Contexts

- User-Service (Identity & Fayda/Passport Verification)
 - Listing-Service (Marketplace Inventory & Multi-Room Hotel Units)
 - Booking-Service (Atomic Inventory Locking & Booking Stepper State Engine)
 - Payment-Service (EasyService Wallet Balance & Telebirr/CBE Transfers)
 - Notification-Service (Real-time Event Alerts & Pre-Booking Reminders)
 - Wheel-Promo-Service (Daily Spin Cost Limits & Unique Single-Use Promo Generator)
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Slide 4: Domain Model: Provider vs. Bookable Units

Hierarchical Property Inventory Model

- Root Provider/Business (e.g. Kuriftu Resort & Spa, Skylight Hotel, Habesha Kemis)

- Listing Metadata: Location, Amenities, Ratings, Photos & Policies
 - Bookable Units / Options: Standard Room (3,500 ETB), Deluxe Room (5,000 ETB), Family Villa (10,000 ETB)
 - Car Fleet Units: Toyota RAV4, Hyundai Elantra, Land Cruiser V8 (Daily Rates)
 - Store Products: Yirgacheffe Coffee, Silk Kemis, Leather Briefcase (Item Units)
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Slide 5: Hotels & Accommodation Marketplace Engine

Kuriftu, Skylight & Premium Resort Booking Flow

- Discovery Page: Filter by Price, Star Rating, Property Type & Amenities
 - Property Detail Page: Image Gallery, Location Coordinates & Full Specs
 - Room Selection Matrix: Direct room selection with real-time room availability counts
 - Date Pickers: Interactive Check-in & Check-out date range selection
 - Nights Multiplier: Automated subtotal calculation based on stay duration
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Slide 6: Car Rental & Mobility Marketplace Engine

Vehicle Fleet Specs & Rental Duration Engine

- Fleet Inventory: SUV, Sedan, Luxury & Off-Road categories
 - Vehicle Specs: Seats, Transmission, Fuel Type, Driver Option & Unlimited KM
 - Rental Date Range: Pickup Date and Return Date selector
 - Day Multiplier: Automatic rental duration calculation (Daily Rate × Days × Quantity)
 - Instant Validation: Return date must strictly be after pickup date
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Slide 7: Cultural Products, Crafts & Events Portal

Ethiopian Heritage Kemis, Leather Goods & Ticket Booking

- Authentic Local Stores: Habesha Heritage Kemis, Entoto Spices, Axum Crafts
 - Item Variants: Specific sizes, silk borders, and material specifications
 - Concerts & Cultural Events: Great Ethiopian Run, Jan Meda Festival tickets
 - Preferred Date Picker: Schedule delivery or event attendance date
 - Direct Provider Settlement: Integrated commission split engine
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Slide 8: Customer Identity & Fayda Verification

Zero-Trust Security & National ID Integration

- AAU Academic Domain SSO: Restricted registration for verified student/faculty credentials
 - Fayda National Digital ID Integration: Verification badge (FAYDA_VERIFIED)
 - Passport Identity Status: Level-2 identity trust scores
 - Stateless JWT Security: RSA-256 signed bearer tokens
 - Role-Based Access Control: CUSTOMER, PROVIDER, SYSTEM_ADMIN roles
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Slide 9: Reactive EasyService Wallet System

Real-time Balance Management & Atomic Deductions

- Simulated Wallet Store: \$currentUser reactive state management
 - Instant Balance Check: Blocks transaction if balance < total payable
 - Atomic Deduction: Mutates user balance instantly across UI navbar & passport
 - Alternative Payment Channels: Integrated Telebirr, CBE Birr & Card options
 - Instant Refund Engine: Wallet credits returned upon booking cancellation
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Slide 10: Spin-the-Wheel Promotion Engine

Tiered Spin Costs & Unique Single-Use Promo Codes

- Tiered Cost Rules: Spins 1-3 (FREE), Spins 4-9 (50 ETB), Spins 10-15 (100 ETB)
 - Wallet Protection: Deducts spin cost directly from EasyService wallet balance
 - Randomized Code Generator: Generates unique code per spin (e.g. WIN10-8492)
 - Central Code Registry: Registers code in activePromoCodes store to prevent reuse
 - Clipboard Copy Action: 1-Click copy button with text fallback
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Slide 11: Hot Deals & Dynamic Pricing Engine

Pre-applied Promos & Flash Sale Bookings

- Marketing Banner Integration: Direct 1-click 'Book Deal →' triggers
 - Pre-applied Promo Deep Linking: Automatically loads promo codes (SUMMER20, ETHIO30)
 - Dynamic Savings Calculation: Displays percentage discount & ETB amount saved
 - Service Fee Inclusion: Transparent 5% platform service fee breakdown
 - Real-time Subtotal Formula: $(\text{Unit Price} \times \text{Quantity} \times \text{Duration} - \text{Discount}) + \text{Fee}$
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Slide 12: End-to-End Booking Stepper Architecture

5-Step Transactional Booking Flow

- Step 1 (Select): Quantity, Dates (Check-in/out), and Promo Code input
 - Step 2 (Details): Customer Name, Email, Phone & Identity Status verification
 - Step 3 (Review): Comprehensive line-item summary and cancellation policy
 - Step 4 (Payment): Payment method selection (Wallet, Telebirr, CBE) & charge execution
 - Step 5 (Confirm): Green receipt badge, reference ID, schedule dates & wallet update
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Slide 13: Date Range Pickers & Schedule Calculation

Native Date Controls & Duration Engine

- Native HTML5 Date Pickers: Minimum date bounded to current day (getTodayStr)
 - Date Difference Engine: $\text{Math.ceil}((\text{end} - \text{start}) / 86,400,000)$
 - Category-Aware Labels: Check-in/out for Hotels, Pickup/Return for Cars
 - Date Range Badge: Visual summary badge displaying computed duration
 - Confirmed Schedule Payload: Persists dates to bookingResult and store history
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Slide 14: Microservices Technical Architecture

Spring Boot 3.2, REST & Messaging Mesh

- Framework: Java 21 LTS with Spring Boot 3.2 Cloud-Native stack
 - API Gateway: Spring Cloud Gateway for routing, rate limiting & SSL termination
 - Service Discovery: Netflix Eureka Service Registry
 - Inter-Service Sync: RESTful HTTP/2 APIs & gRPC binary protocol
 - Async Messaging: Apache Kafka event streams for notifications and Saga triggers
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Slide 15: Database Schema & Data Isolation

Database per Service Pattern Implementation

- Isolated Storage: Dedicated PostgreSQL schema per microservice boundary
- User DB: Customer profiles, credentials & identity verification flags
- Listing DB: Provider properties, room units, specs & inventory counters

- Booking DB: Reservation ledger, stepper status & timestamp audit logs
 - Performance Indexing: B-Tree indexes on listing_id, customer_id & status
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Slide 16: Frontend Architecture & UX Design

Svelte 4 Reactive Store & Custom CSS Token System

- UI Framework: Svelte 4 compiler for zero-virtual-DOM performance
 - State Management: Reactive stores (authStore.js, bookingStore.js)
 - Visual Aesthetic: Modern dark theme, glassmorphism & gold accents (#d4af37)
 - Component Library: Modular components (BookingModal, SpinWheelModal, Navbar)
 - Accessibility (A11y): ARIA roles, high-contrast text & modal keyboard focus
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Slide 17: API Specifications & Endpoints

OpenAPI 3.0 Standardized Service Contracts

- GET /api/listings - Fetch 28 curated Ethiopian marketplace business listings
 - POST /api/bookings?customerId=... - Execute atomic reservation transaction
 - POST /api/wallet/topup - Reactive wallet credit deposit
 - POST /api/spin/wheel - Execute promotional spin & register unique code
 - Swagger UI: Live API documentation interactive sandbox at /swagger-ui.html
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Slide 18: Quality Assurance & Test Suite

Testing Pyramid & Defect Metric Analysis

- Unit Testing: JUnit 5 & Mockito testing core domain business rules
 - Contract Testing: Pact framework verifying microservice API provider/consumer contracts
 - Integration Testing: Testcontainers PostgreSQL & Kafka containerized integration tests
 - End-to-End Testing: Playwright automated headless browser test flows
 - Defect Zero Guarantee: Built-in fail-safe fallbacks for offline demo resiliency
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Slide 19: Deployment & CI/CD Pipeline

Docker Containerization & Cloud Hosting Strategy

- Containerization: Multi-stage Dockerfiles producing distroless Alpine images
 - Orchestration: Kubernetes manifests (Deployment, Service, Ingress, HPA)
 - Continuous Integration: GitHub Actions workflow building and testing images
 - Cloud Deployment: Railway / Vercel containerized deployment
 - Observability: Prometheus metrics scraper & Grafana dashboard visualizer
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Slide 20: Conclusion & Project Roadmap

Summary of Accomplishments & Future Extensions

- Accomplishment: Fully functional, hardened Ethiopian Service Marketplace platform
 - Wallet Integrity: Reactive, fail-safe wallet payments and spin wheel charges
 - Promo Engine: Unique single-use promo code generation and dynamic discount parsing
 - Booking Flow: Seamless check-in/out date pickers and night multipliers
 - Next Phase: Real-world Spring Boot backend REST integration & Telebirr Production API
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